

Per-variant tissue assignment by a sequence-to-function model is limited by effect detectability more than by assay coverage

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Abstract

Sequence-to-function models predict tissue-resolved regulatory activity from DNA, so they can be asked in which tissue an individual non-coding risk variant acts. We tested whether AlphaGenome recovers trait-tissue relationships established by orthogonal evidence, scoring fine-mapped variants from 20 GWAS of 19 complex traits across 11 tissue groups against a shared background of 318 variants. The pre-specified expected tissue ranked first for 11 of 20 trait-GWAS pairs ($P = 2.7 \times 10^{-7}$), within the top three for 16, and was significantly enriched for 14. Recovery did not track how well the expected tissue is measured: track count was uncorrelated with enrichment ($\rho = -0.01$) or rank ($\rho = +0.25$), although the test can exclude only a strong coverage effect. Enrichment did track a tissue-agnostic effect visibility ($\rho = +0.81$), but largely by construction, because predicted histone effects are shared across tissues (mean inter-tissue correlation 0.80); visibility did not predict the expected tissue’s rank ($\rho = +0.31$, $P = 0.19$), and at the variant level tissue information was modest and concentrated in the most visible variants (within-trait odds ratio 1.45). Matched against LDSC-SEG on the same 16 GWAS and tissue definitions, the aggregate method was ahead (rank-1 recovery 10 versus 8; top three 15 versus 13). Its advantage was concentrated where per-variant effects are smallest: it placed brain first for both brain-expected traits ($P = 0.002$ and 2.4×10^{-4}) and the islet first for type 2 diabetes, where AlphaGenome ranked them third, third and 8th, whereas AlphaGenome recovered artery for coronary artery disease where LDSC-SEG found no significant tissue. On 2 of the 4 traits defeating both, the methods chose the same wrong tissue, implicating the labels. A null per-variant tissue assignment therefore more often reflects undetected effects than absent tissue involvement, and the tissue-specific component of predicted effects is small relative to the shared one.

1 Introduction

Most risk variants for common disease are non-coding (Maurano et al., 2012; Finucane et al., 2015). Interpreting them requires knowing not only the target gene but the cell type in which the variant acts, since a regulatory element is functional only in tissues that use it. The established approach is statistical: partition trait heritability across tissue-specific annotations and test for enrichment (Finucane et al., 2015, 2018). These methods consume whole GWAS summary statistics and require matched functional annotations for each tissue.

Sequence-to-function models offer a different route. Enformer (Avsec et al., 2021), Borzoi (Linder et al., 2025) and AlphaGenome (Avsec et al., 2026) predict functional genomic tracks directly

from sequence. AlphaGenome accepts one megabase of DNA and predicts thousands of tracks spanning expression, accessibility, histone modification, transcription factor binding, contact maps and splicing at up to base-pair resolution. Because one model emits predictions for many tissues at once, contrasting reference and alternate alleles yields a per-tissue effect estimate for a *single* variant—something heritability partitioning cannot provide, as it operates on genome-wide aggregates.

Whether those per-variant tissue assignments are trustworthy at the level of a whole trait is largely untested. This matters practically: a null prediction is easily misread as evidence that a tissue is *not* involved, when it may instead mean the model failed to detect the variant’s effect in any tissue. Distinguishing these requires a benchmark with known answers, and—if recovery turns out to be partial—an account of what separates success from failure.

We built such a benchmark and used it to adjudicate between two candidate explanations for failure. The *assay-coverage* hypothesis holds that a tissue is recovered when the model has seen enough experimental data for it. The *detectability* hypothesis holds that recovery depends on whether the trait’s variants produce predicted effects large enough to rank tissues at all, independently of which tissue is expected. These make opposite predictions for well-instrumented tissues paired with subtle traits. We find no support for the first and, from a matched comparison with the reference method, direct support for the second, with an important qualification: most of what the model predicts for a variant is shared across tissues, and the tissue-specific remainder is small.

2 Results

2.1 Benchmark design

We assembled 20 trait–GWAS pairs covering 19 traits (Appendix Table 3) whose principal tissue is supported by prior literature, spanning liver, immune tissue, heart, artery, brain, lung and skin, plus type 2 diabetes (T2D). For each trait we took fine-mapped 95% credible sets from the Open Targets Platform (Buniello et al., 2025) and kept the highest-posterior non-coding variant per independent signal, capped at 150 signals per trait. The Platform’s fine-mapping method depends on what it has ingested for a study: SuSiE-inf (Wang et al., 2020) where full summary statistics are available, otherwise PICS applied either to summary statistics or, when none were ingested, to the reported lead variants alone. In our panel 5 studies are SuSiE-based, 10 are PICS on summary statistics and 5 are PICS around reported top hits (Appendix Table 3). For the last group the highest-posterior variant is in practice the reported lead SNP, so for those traits the benchmark scores GWAS lead variants rather than fine-mapped causal candidates.

Four datasets in the panel as first assembled were replaced after an internal review of the first complete analysis: two Million Veteran Program PheCode GWAS with fewer than four thousand cases (ulcerative colitis, and type 1 diabetes with ophthalmic manifestations), and the body mass index and cognition datasets, whose summary statistics are not distributed through the GWAS Catalog and which therefore could not enter the matched comparison of Section 2.5. Each was replaced by the largest GWAS of the same trait with summary statistics ingested by Open Targets and available from the Catalog. The expected-tissue labels were not changed. The superseded datasets and their results are reported in Appendix Table 5.

Two points about provenance, because the benchmark is easy to misread. First, the Open Targets Platform supplies *only the variant list*. It makes no tissue predictions, and its locus-to-gene scores are not used anywhere in the primary analysis, which is histone-based and involves no gene assignment. Every tissue ranking reported here is AlphaGenome’s alone. Second, the expected tissue for each trait is a label we assigned from prior literature before any scoring, not a

quantity derived from data in this study. The benchmark therefore measures agreement between AlphaGenome and an external literature consensus, and the labels are the weakest link in it: they are judgements, and Section 2.5 shows two of them are contradicted by both methods.

Each variant was scored with AlphaGenome over a 1 Mbp window. Histone modification tracks are the primary readout: the RNA-seq scorer requires choosing a target gene per variant, and any such rule differs systematically between trait variants (which have a nominated gene) and background variants (which do not), biasing the comparison. Histone centre-mask scores involve no such choice. We used the five marks present in all tissue groups (H3K27ac, H3K36me3, H3K4me1, H3K4me3, H3K9ac).

Absolute calibration requires a null, so we generated a single shared background of 318 single-nucleotide variants placed 50 kbp to 400 kbp from T2D and LDL-cholesterol signals on the same chromosomes, with reference alleles from GRCh38. The background is not matched to trait variants on allele frequency, linkage disequilibrium or annotation, so enrichment over it partly reflects that GWAS variants sit in regulatory sequence at all; the rank of the expected tissue among the eleven is the measure that speaks to tissue identity. Every per-tissue score is standardised against this background, so each trait is scored independently.

2.2 Recovery is well above chance but incomplete

The expected tissue ranked first for 11 of 20 trait–GWAS pairs against 1.8 expected by chance ($P = 2.7 \times 10^{-7}$), and within the top three for 16 against 5.5 ($P = 1.4 \times 10^{-6}$). Type 2 diabetes contributes two pairs; counting it once leaves the conclusion unchanged ($P = 1.3 \times 10^{-7}$ or 1.6×10^{-6} depending on which dataset is kept). A null that resamples background variant sets of the same size, and so allows for unequal tail behaviour across tissues, gives an expected 1.7 rank-1 recoveries rather than 1.8. Judged by enrichment rather than rank, the expected tissue was significantly above background for 14 pairs (BH $q < 0.05$). Table 1 gives the full ranking with, for each pair, the proportion of variant resamples in which the expected tissue stays first; 3 of the 11 rank-1 recoveries have bootstrap support below one half. Figure 1 shows the trait-by-tissue matrix.

Liver traits recovered liver (alanine aminotransferase $z = 1.33$), cardiac traits recovered heart, and platelet count, ulcerative colitis, type 1 diabetes and the Crohn’s disease medication proxy recovered immune tissue. Rank and enrichment do not always agree: atrial fibrillation and the larger T2D dataset rank the expected tissue first without its enrichment reaching significance, and when every tissue sits near zero the ordering is noise. We therefore report both. The superseded datasets make the same point from the other side: the two small PheCode GWAS in the original panel showed no enrichment in any tissue, and their large-consortium replacements both recover immune tissue with significant enrichment (Appendix Table 5). The benchmark tests the input variant set as much as the model.

2.3 Assay depth does not explain which traits are recovered

If recovery were limited by how well a tissue is measured, the expected tissue’s enrichment should rise with the number of tracks available for it. We find no evidence that it does. Across trait–GWAS pairs, enrichment was uncorrelated with the expected tissue’s histone track count ($\rho = -0.01$, $P = 0.97$) and with its chromatin accessibility track count ($\rho = +0.07$, $P = 0.76$); both point estimates are close to zero. The same holds for rank ($\rho = +0.25$, $P = 0.29$; accessibility $\rho = +0.31$, $P = 0.18$). This is a low-powered test: track count takes only eight distinct values across the 20 pairs, because traits that share an expected tissue share a count, and the 95% confidence interval on the first correlation runs from -0.45 to +0.43. The data exclude a strong coverage dependence,

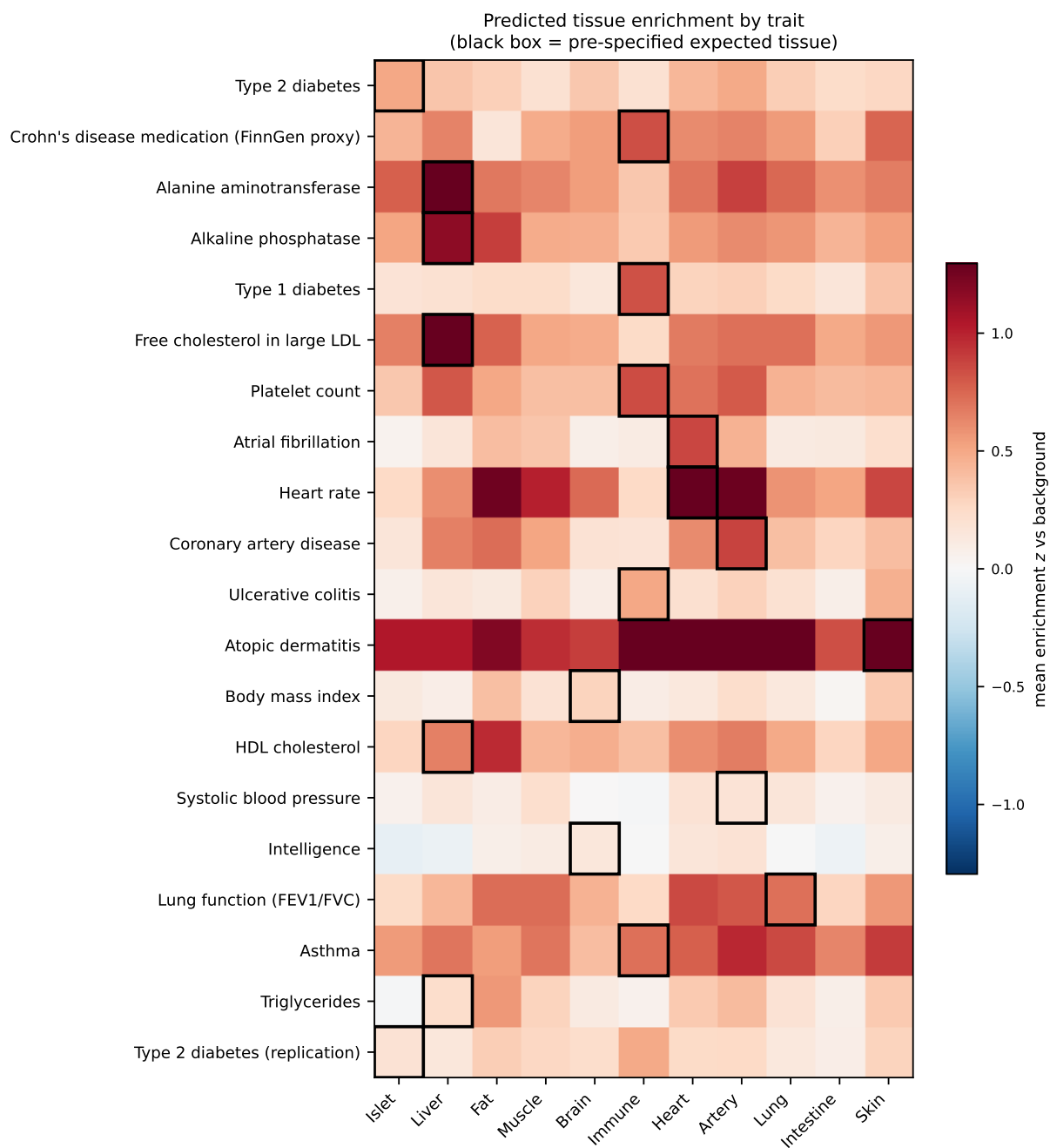


Figure 1: Mean predicted enrichment over background for each trait across 11 tissue groups, ordered by the rank achieved by the pre-specified expected tissue (black box). Traits at the bottom show little signal in any tissue.

Table 1: Trait–tissue recovery. **Rank** is the position of the expected tissue among 11 tissues, ordered by mean enrichment over background. **Boot.** is the proportion of 2,000 resamples of the trait’s variants in which the expected tissue ranks first. z_{exp} is the expected tissue’s enrichment; P is a one-sided Mann–Whitney test of that tissue against background. Type 2 diabetes appears twice, once per GWAS.

Trait	Expected	n	Rank	Boot.	Top-ranked	z_{exp}	P
Heart rate	Heart	150	1	0.65	Heart	1.41	1.0e-03
Alanine aminotransferase	Liver	150	1	0.93	Liver	1.33	3.3e-05
Platelet count	Immune	150	1	0.45	Immune	0.85	2.3e-08
Crohn’s disease medication (FinnGen proxy)	Immune	50	1	0.48	Immune	0.83	2.7e-03
Free cholesterol in large LDL	Liver	111	1	0.94	Liver	1.29	2.4e-03
Alkaline phosphatase	Liver	150	1	0.76	Liver	1.16	9.1e-03
Coronary artery disease	Artery	150	1	0.79	Artery	0.88	4.1e-03
Type 1 diabetes	Immune	66	1	0.92	Immune	0.82	0.03
Type 2 diabetes	Islet	150	1	0.44	Islet	0.50	0.07
Ulcerative colitis	Immune	150	1	0.59	Immune	0.50	4.2e-03
Atrial fibrillation	Heart	150	1	0.96	Heart	0.87	0.12
Atopic dermatitis	Skin	85	2	0.04	Immune	1.77	1.6e-03
HDL cholesterol	Liver	150	3	0.08	Fat	0.65	9.8e-05
Body mass index	Brain	150	3	0.17	Fat	0.29	0.19
Systolic blood pressure	Artery	148	3	0.05	Muscle	0.17	0.34
Intelligence	Brain	150	3	0.25	Artery	0.15	0.87
Asthma	Immune	122	5	0.11	Artery	0.72	5.4e-05
Lung function (FEV1/FVC)	Lung	150	5	0.15	Heart	0.71	2.2e-04
Triglycerides	Liver	109	6	0.00	Fat	0.22	0.02
Type 2 diabetes (replication)	Islet	118	8	0.05	Immune	0.18	0.17

not a moderate one.

The extremes make the point directly. Brain is among the best-instrumented groups (63 histone tracks, 45 of them among the five marks scored here, and 19 accessibility tracks), yet both brain-expected traits fail: body mass index ranks the brain 3rd ($z = 0.29$, $P = 0.19$) and intelligence 3rd ($z = 0.15$, $P = 0.87$). The pancreatic islet is the sparsest group in the panel (5 histone tracks, 1 accessibility track, no ATAC), yet T2D ranks the islet first in the larger GWAS. That recovery is weak (non-significant, bootstrap support 0.44) and is not reproduced in the second T2D dataset (Section 2.6), so the islet example shows only that sparse coverage does not preclude recovery, not that coverage is irrelevant.

2.4 Effect detectability explains significance; its link to tissue identity is weaker

We defined a tissue-agnostic measure of *effect visibility*: the mean predicted effect of a trait’s variants across all tissue-assigned histone tracks, standardised against the same background. This quantity does not reference the expected tissue.

Visibility tracks expected-tissue enrichment closely ($\rho = +0.81$, $P = 1.6 \times 10^{-5}$). Traits whose expected tissue was significantly enriched had median visibility 0.69, against 0.27 for the rest ($P = 4.9 \times 10^{-4}$). Much of this association is, however, expected by construction. Predicted histone effects are strongly shared across tissues: over the 2,609 scored variants the mean pairwise inter-tissue correlation of standardised effects is 0.80 (range 0.69–0.91), and a single component explains 76% of their variance. Visibility therefore correlates with the enrichment of *whichever* tissue ranks first ($\rho = +0.85$, $P = 1.8 \times 10^{-6}$) as strongly as with the expected tissue’s. A trait whose variants have large predicted effects will show significant enrichment of its expected tissue almost regardless of whether the model has located the tissue correctly.

The scale-free question is whether visibility predicts the expected tissue’s *rank*. Across trait–GWAS pairs it does not, significantly: $\rho = +0.31$ ($P = 0.19$) with rank, $\rho = +0.16$ ($P = 0.49$) with bootstrap support for rank 1, and median visibility 0.68 for rank-1 pairs against 0.36 for the rest ($P = 0.11$). Of the ten most visible pairs, 6 rank the expected tissue first, as do 5 of the ten least visible.

At the level of individual variants the picture is sharper. For each scored variant we asked whether the expected tissue ranked first among the eleven, against a null calibrated on background variants (which is not uniform: immune tissue ranks first for 30.5% of background variants, liver for 5.0%). Overall the expected tissue ranked first for 17.4% of variants against 11.7% expected, a 1.48-fold excess. The excess is concentrated in the most visible variants: 1.78-fold in the top visibility quartile ($P = 8.3 \times 10^{-12}$) against 1.50, 1.29 and 1.34-fold in the lower three. Within traits, variants in a trait’s own top visibility quartile were 1.45 times as likely to rank the expected tissue first as the trait’s other variants (logistic regression with trait fixed effects, 95% CI 1.15–1.84, $P = 1.9 \times 10^{-3}$; 1.10 per standard deviation of visibility, $P = 0.04$). Tissue information in the predictions is therefore real but modest, and it is carried mainly by the variants with the largest predicted effects. Where a trait has few such variants, its tissue ranking rests on a handful of loci, which is why several rank-1 recoveries have low bootstrap support.

2.5 Head-to-head against LDSC-SEG

The established method for this question is stratified LD score regression applied to specifically expressed genes (LDSC-SEG) (Finucane et al., 2018). Unlike AlphaGenome it never scores an individual variant: it asks whether a trait’s heritability, aggregated across all SNPs, is concentrated near the genes a tissue expresses distinctively. If the detectability account is right, that difference should matter precisely where per-variant effects are too small to detect.

We ran it matched: the 16 traits for which harmonised summary statistics were available from the GWAS Catalog, the same GWAS datasets as the AlphaGenome benchmark, and the same 11 tissue definitions. We built the LDSC-SEG annotations from GTEx v8 median expression ourselves so that both methods partition the genome into identical tissue groups, rather than comparing against the 205-annotation taxonomy of the published analysis. Table 2 gives the result.

LDSC-SEG is ahead. The expected tissue ranked first for 10 of 16 traits under LDSC-SEG and 8 of 16 under AlphaGenome, both far above the 1.5 expected by chance ($P = 3.1 \times 10^{-5}$ for AlphaGenome and smaller for LDSC-SEG). LDSC-SEG also leads on top-three recovery (15 versus 13) and on rank-1 recoveries that are also significant for the expected tissue (10 versus 7). The margin is consistent across measures but not large, and the two methods fail on different traits.

The failures are *complementary*, and where the methods diverge the direction is informative. For the two brain-expected traits LDSC-SEG places brain first with strong significance (body mass index $P = 0.002$; intelligence $P = 2.4 \times 10^{-4}$) where AlphaGenome ranks it 3rd and 3rd, behind fat and artery. For type 2 diabetes LDSC-SEG ranks the islet first ($P = 0.032$) where AlphaGenome ranks it 8th. In all three cases the aggregate method recovers tissue biology the literature independently supports (Gaulton et al., 2010; Pasquali et al., 2014; Chiou et al., 2021) from GWAS whose individual fine-mapped variants AlphaGenome cannot resolve. These are the traits with the smallest per-variant effects, and this is the head-to-head test the detectability account required. For coronary artery disease the ordering reverses: AlphaGenome ranks artery first with significant enrichment ($q = 0.008$) while LDSC-SEG ranks it 3rd behind lung and fat, with no tissue reaching significance (best $P = 0.05$). For type 1 diabetes both methods find immune signal, but LDSC-SEG places lung marginally ahead of immune tissue (immune $P = 0.016$), plausibly because bulk lung carries a substantial immune-cell expression signature. Whatever separates success from

failure, it is not that one method is uniformly better resolved.

The two also fail *together* informatively. Both methods miss on 4 traits, and on 2 of those the methods independently choose the same wrong tissue: both rank fat above liver for HDL cholesterol, and both put immune tissue first for atopic dermatitis, AlphaGenome with its strongest enrichment in the panel ($z = 2.47$) and LDSC-SEG with $P = 2.4 \times 10^{-5}$. When two methods sharing almost no machinery agree against our label, the label is the likelier error. This is the strongest evidence here that the pre-specified expected tissues, not the methods, limit the benchmark.

Table 2: Matched head-to-head. Both methods were run on the same 16 traits, the same GWAS datasets and the same 11 tissue definitions. **Rank** is the position of the pre-specified expected tissue. Bold marks a rank-1 recovery.

Trait	Expected	AlphaGenome		LDSC-SEG	
		Rank	Top	Rank	Top
Free cholesterol in large LDL	Liver	1	Liver	1	Liver
Alanine aminotransferase	Liver	1	Liver	1	Liver
Alkaline phosphatase	Liver	1	Liver	1	Liver
Ulcerative colitis	Immune	1	Immune	1	Immune
Atrial fibrillation	Heart	1	Heart	1	Heart
Heart rate	Heart	1	Heart	1	Heart
Body mass index	Brain	3	Fat	1	Brain
Intelligence	Brain	3	Artery	1	Brain
Asthma	Immune	5	Artery	1	Immune
Type 2 diabetes (replication)	Islet	8	Immune	1	Islet
Type 1 diabetes	Immune	1	Immune	2	Lung
HDL cholesterol	Liver	3	Fat	2	Fat
Systolic blood pressure	Artery	3	Muscle	2	Fat
Coronary artery disease	Artery	1	Artery	3	Lung
Lung function (FEV1/FVC)	Lung	5	Heart	3	Artery
Atopic dermatitis	Skin	2	Immune	7	Immune
<i>rank-1 recovery</i>		8/16		10/16	
<i>top-3 recovery</i>		13/16		15/16	

2.6 Type 2 diabetes is unstable, not null

T2D behaves as the detectability account predicts, and its behaviour is a caution against over-reading single-dataset results. In the larger GWAS (Mahajan et al., 2018) the islet ranks first, consistent with the substantial literature localising T2D risk to islet regulatory elements (Gaulton et al., 2010; Pasquali et al., 2014; Greenwald et al., 2019; Chiou et al., 2021)—but the enrichment is modest and does not reach significance ($z = 0.50$, $P = 0.072$). In an independent GWAS (Xue et al., 2018) the islet ranks 8th ($z = 0.18$).

The two variant sets share only 15 variants. They differ in GWAS and in fine-mapping method (PICS around reported lead variants for the larger dataset, SuSiE-inf on summary statistics for the other), so they are near-disjoint samples of the same underlying signal, and the disagreement is partly a statement about how much the scored variant depends on those choices. The matched LDSC-SEG run (Section 2.5) resolves what AlphaGenome alone could not: on the same Xue dataset where AlphaGenome ranks the islet 8th, LDSC-SEG ranks it *first* ($P = 0.032$, nominal, with bulk pancreas standing in for islet). The islet signal is present in that GWAS; AlphaGenome does not detect it in the fine-mapped variants. That upgrades our reading of T2D from “uninformative”

to a specific negative—per-variant prediction misses islet enrichment that aggregate heritability partitioning recovers from the same study—and is consistent with the instability across the two T2D datasets reflecting the model’s detection threshold rather than a genuine absence of islet involvement. The two brain-expected traits show the same signature without the instability: one dataset each, brain first under LDSC-SEG and third under AlphaGenome (Section 2.5).

One caveat compounds this. The larger dataset used islet epigenome maps in its own fine mapping (Mahajan et al., 2018). Open Targets has not ingested its summary statistics, so the credible sets we used are PICS sets built around the reported index variants. The annotation-informed credible sets of the original study therefore do not enter, but the index variants come from a study whose signal definition was islet-aware, and any residual influence would bias *towards* islet variants—so the islet-first result in that dataset should be treated as an upper bound, not a confirmation.

3 Discussion

Our benchmark supports a narrow claim with practical consequences. AlphaGenome’s per-variant tissue assignments recover independently established trait–tissue relationships far above chance, and can be used to generate tissue hypotheses for individual non-coding variants. But significant recovery is conditional on the model predicting a detectable effect for those variants, and 6 of our 20 trait–GWAS pairs fall below that threshold. For those the tissue ranking rests on a handful of loci and carries little information, and crucially, nothing in the output distinguishes this state from a confident negative. A second point follows from the structure of the predictions themselves: the tissue-specific part of a predicted variant effect is small next to the part shared across tissues, so a per-variant tissue call is a weak signal even when the variant is visible, and it should be aggregated over loci before it is trusted.

That the limiting factor looks like detectability rather than assay coverage is the more useful finding, because it redirects effort. Adding chromatin accessibility assays for under-measured tissues would be a natural response to a coverage-limited failure; our results suggest it would not, by itself, rescue the traits that fail here, since the best-instrumented tissue in the panel hosts two of the clearest failures. The relevant axis is instead the size of the regulatory effects a trait’s variants actually have. The matched comparison makes this concrete: for body mass index, intelligence and type 2 diabetes, the three misses whose variants are among the least visible to the model, LDSC-SEG recovers the expected tissue from the same GWAS, so the tissue signal is present in the data and absent only from the per-variant predictions. Highly polygenic traits—body mass index, cognition, T2D—are those whose individual variants have small effects, and they are among the ones that fail, although the pattern is not clean: platelet count is at least as polygenic and is recovered strongly. Two alternative readings of low visibility cannot be excluded by this design. The scored variant may not be the causal one, since 5 of our 20 variant sets are lead SNPs rather than fine-mapped candidates, and visibility rises weakly with posterior inclusion probability at the variant level ($\rho = +0.15$, $P = 3.8 \times 10^{-15}$); and the signal itself may be marginal, since visibility also rises with GWAS significance ($\rho = +0.04$ per variant, $P = 0.03$) and the two datasets removed from the original panel after review, both from GWAS with fewer than four thousand cases, were the least visible of all (Appendix Table 5). Beyond these, the brain failures may reflect that the relevant cell types and developmental stages are poorly represented by bulk adult brain tracks. The detectability reading is consistent with independent reports that current sequence models capture promoter-proximal determinants well but distal enhancer effects poorly (Karollus et al., 2023), and explain inter-individual expression variation weakly (Sasse et al., 2023; Huang et al., 2023).

For T2D specifically, the biological picture in the literature is not a single tissue in any case: T2D loci partition into mechanistically distinct clusters, only a subset of which are beta-cell mediated (Udler et al., 2018; Lotta et al., 2017). A single expected tissue label is a simplification for this trait in a way it is not for alanine aminotransferase, and this may contribute to its low visibility.

What would sharpen this

Three follow-ups would strengthen or overturn these conclusions. First, complete the matched comparison: the FinnGen Crohn’s disease proxy, the DIAGRAM T2D dataset, platelet count and triglycerides have public summary statistics outside the GWAS Catalog and were not run through LDSC-SEG. Second, aggregate over full credible sets weighted by posterior probability instead of taking the top variant, which would quantify how much of the failure is fine-mapping instability—the near-disjoint T2D variant sets suggest this is substantial. Third, test the detectability account prospectively: rank traits by visibility before looking at tissue assignments, and check that the predicted recovery ordering holds on a held-out trait set. Supplying external islet accessibility data (Chiou et al., 2021) remains worth doing, but our results reframe it as a test of the coverage hypothesis rather than an expected fix.

4 Limitations

1. **One variant per signal.** We scored the highest-posterior variant per credible set; the median 95% credible set in the SuSiE-fine-mapped T2D replication data contains 5 variants, so the causal variant is often a different member.
2. **Bulk tissue dilutes cell types.** Most tracks are bulk tissue, and attenuation of cell-type-specific effects is not equal across groups.
3. **Expected tissues are author-assigned judgements.** This is the benchmark’s principal weakness. Labels were fixed before scoring and drawn from prior literature, but they are not derived from data and several traits have more than one defensible principal tissue (atopic dermatitis: skin and immune; coronary artery disease: artery and liver; alkaline phosphatase: liver and bone). The LDSC-SEG comparison contradicts two of the twenty (Section 2.5); others may be equally contestable. We report rank rather than a binary hit partly for this reason. A benchmark with labels derived from an independent quantitative method, rather than assigned by hand, would be stronger.
4. **Proxy phenotypes.** The Crohn’s disease entry is a FinnGen first-line-medication endpoint with 106,320 cases, far broader than Crohn’s disease, and heart rate is a Million Veteran Program PheWAS trait. Case counts and definitions are given in Table 3. Two PheCode datasets with fewer than four thousand cases were removed from the panel after review (Appendix Table 5); their credible sets were almost all singletons at marginal significance, and their failures were failures of the input rather than of the model.
5. **Rank is unstable when enrichment is flat.** Where all tissues sit near zero, the top-ranked tissue is arbitrary; this is why we report enrichment and significance alongside rank.
6. **Skewed effect distributions.** Median per-variant scores are near or below zero for every trait; enrichment is driven by a minority of loci. Means are appropriate for enrichment but describe the tail.

7. **Four traits lack a matched comparison.** The FinnGen Crohn’s disease proxy, the DI-AGRAM T2D dataset, platelet count and triglycerides have summary statistics outside the GWAS Catalog and were not run through LDSC-SEG.
8. **The LDSC-SEG islet annotation is bulk pancreas.** GTEx has no islet tissue, so on the LDSC-SEG side the islet group is whole pancreas—the same cell-type dilution affecting the AlphaGenome side, and a conservative bias for the diabetes result.
9. **Some LDSC-SEG ranks rest on non-significant coefficients.** Where no tissue reaches significance the ordering is arbitrary, exactly as for AlphaGenome; coronary artery disease is the clearest case.
10. **LDSC required patching to run.** The Python 3 port used here contained defects that silently altered results: chromosome-split LD scores were read for a single chromosome instead of 22, and cell-type P -values were two-tailed where the published method is one-tailed. Both were corrected and validated (Methods).
11. **Heterogeneous fine-mapping.** Credible sets come from three regimes (SuSiE-inf, PICS on summary statistics, PICS around reported lead variants), and for 5 of 20 datasets the scored variant is effectively the GWAS lead SNP. Trait-to-trait differences in visibility therefore partly reflect how well the scored variant stands in for the causal one.
12. **Unmatched background.** The background is a single set of 318 SNVs near T2D and LDL loci, not matched on allele frequency, LD or annotation. Enrichment over it mixes tissue identity with the generic regulatory character of GWAS variants; rank and the calibrated per-variant analysis are the tissue-specific readouts.
13. **Low power for the coverage test.** With 20 pairs and eight distinct expected tissues, the correlation between track count and recovery can exclude only a strong coverage effect.
14. **Single model, single readout.** One model, histone tracks only. Whether Enformer or Borzoi show the same detectability dependence is untested.
15. **Predictions, not measurements.** Every quantity is a model output; none is experimentally validated here.

5 Methods

Variant selection. Fine-mapped credible sets were retrieved from the Open Targets Platform GraphQL API (Buniello et al., 2025). For each trait we identified GWAS studies mapped to the trait’s ontology term, selected the study with the most credible sets, and retrieved up to 400. Within each set we kept members flagged as belonging to the 95% credible set and took the variant with highest posterior inclusion probability. Signals whose highest-posterior variant had a coding most-severe consequence (missense, stop, frameshift, synonymous, canonical splice, in-frame) were dropped rather than replaced by the next non-coding member. Remaining variants were ordered by signal significance and capped at 150 per trait. Fine-mapping method and confidence for each study were retrieved from the same API and are listed in Table 3. Replacement datasets (Section 2.1) were chosen as the largest GWAS of the trait with summary statistics ingested by Open Targets and harmonised summary statistics on the GWAS Catalog FTP site; the superseded datasets were scored identically and are reported in Appendix C.

Background construction. For each non-coding T2D and LDL-cholesterol signal we drew two positions offset by 50 kbp to 400 kbp in either direction on the same chromosome, and kept 320 at random. Reference bases were retrieved in batch from the Ensembl GRCh38 REST API and each position converted to a transition SNV; 318 passed retrieval and scoring. The same background serves every trait.

Scoring. AlphaGenome (alphagenome Python client) was queried with `score_variant` over `SEQUENCE_LENGTH_1MB` intervals centred on each variant, using the recommended `CHIP_HISTONE` centre-mask and `RNA_SEQ` gene-mask scorers. Histone scores form the primary analysis.

Tissue grouping. Output metadata was grouped into 11 tissue classes by biosample ontology name and GTEx tissue label (GTEx Consortium, 2020; ENCODE Project Consortium, 2020), restricted to histone marks present in all groups.

Statistics. For tissue g and variant v the raw score is the mean absolute predicted effect over the tracks of g , standardised as $z = (x - \mu_g^{\text{bg}}) / \sigma_g^{\text{bg}}$. Trait tissue scores are means of z over variants. Enrichment used one-sided Mann–Whitney U against background with Benjamini–Hochberg correction across the 20 traits. Effect visibility is the same statistic computed over all tissue-assigned histone tracks pooled, standardised against the pooled background. Associations used Spearman correlation. Recovery rates were tested against 1/11 and 3/11 by exact binomial test, treating trait–GWAS pairs as independent; a calibrated chance rate was obtained by resampling background variant sets of size 150 (4,000 draws) and recording which tissue ranked first. Bootstrap support for a rank-1 recovery is the proportion of 2,000 resamples of a trait’s variants (with replacement) in which the expected tissue ranks first. Per-variant analyses use the same standardised scores: the rank of the expected tissue among the 11 groups for each variant, a null rate equal to the frequency with which that tissue ranks first among background variants, quartiles of visibility pooled across traits, and logistic regression of rank-1 status on visibility with trait fixed effects. Inter-tissue correlation is the mean pairwise Spearman correlation of standardised per-variant scores across the 11 groups, and the shared component is the first principal component of that matrix. These analyses are implemented in `scripts/37_review_analyses.py`.

Data availability. All inputs are public: GWAS summary statistics via the accessions in Table 3, fine-mapped credible sets via the Open Targets Platform API, and AlphaGenome predictions via the model’s public API. Analysis scripts, derived score matrices, the LDSC-SEG results and the exact panel definitions are at <https://github.com/ronneldavis/alphagenome-tissue-benchmark>.

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References

Žiga Avsec, Vikram Agarwal, Daniel Visentin, et al. Effective gene expression prediction from sequence by integrating long-range interactions. *Nature Methods*, 18:1196–1203, 2021. doi: 10.1038/s41592-021-01252-x.

- Žiga Avsec et al. Advancing regulatory variant effect prediction with AlphaGenome. *Nature*, 649 (8099):1206–1218, 2026. doi: 10.1038/s41586-025-10014-0. Preprint: bioRxiv 2025.06.25.661532.
- Annalisa Buniello, Daniel Suveges, Carlos Cruz-Castillo, et al. Open targets platform: facilitating therapeutic hypotheses building in drug discovery. *Nucleic Acids Research*, 53(D1):D1467–D1475, 2025. doi: 10.1093/nar/gkae1128.
- Joshua Chiou, Chun Zeng, Zhang Cheng, et al. Single-cell chromatin accessibility identifies pancreatic islet cell type- and state-specific regulatory programs of diabetes risk. *Nature Genetics*, 53:455–466, 2021. doi: 10.1038/s41588-021-00823-0.
- ENCODE Project Consortium. Expanded encyclopaedias of DNA elements in the human and mouse genomes. *Nature*, 583:699–710, 2020. doi: 10.1038/s41586-020-2493-4.
- Hilary K. Finucane, Brendan Bulik-Sullivan, Alexander Gusev, et al. Partitioning heritability by functional annotation using genome-wide association summary statistics. *Nature Genetics*, 47: 1228–1235, 2015. doi: 10.1038/ng.3404.
- Hilary K. Finucane, Yakir A. Reshef, Verner Anttila, et al. Heritability enrichment of specifically expressed genes identifies disease-relevant tissues and cell types. *Nature Genetics*, 50:621–629, 2018. doi: 10.1038/s41588-018-0081-4.
- Kyle J. Gaulton, Takao Nanno, Lorenzo Pasquali, et al. A map of open chromatin in human pancreatic islets. *Nature Genetics*, 42:255–259, 2010. doi: 10.1038/ng.530.
- William W. Greenwald, Joshua Chiou, Jian Yan, et al. Pancreatic islet chromatin accessibility and conformation reveals distal enhancer networks of type 2 diabetes risk. *Nature Communications*, 10:2078, 2019. doi: 10.1038/s41467-019-09975-4.
- GTEx Consortium. The GTEx consortium atlas of genetic regulatory effects across human tissues. *Science*, 369(6509):1318–1330, 2020. doi: 10.1126/science.aaz1776.
- Connie Huang, Richard W. Shuai, Parth Baokar, Ryan Chung, Ruchir Rastogi, Pooja Kathail, and Nilah M. Ioannidis. Personal transcriptome variation is poorly explained by current genomic deep learning models. *Nature Genetics*, 55(12):2056–2059, 2023. doi: 10.1038/s41588-023-01574-w.
- Alexander Karollus, Thomas Mauermeier, and Julien Gagneur. Current sequence-based models capture gene expression determinants in promoters but mostly ignore distal enhancers. *Genome Biology*, 24:56, 2023. doi: 10.1186/s13059-023-02899-9.
- Johannes Linder, Divyanshi Srivastava, Han Yuan, Vikram Agarwal, and David R. Kelley. Predicting RNA-seq coverage from DNA sequence as a unifying model of gene regulation. *Nature Genetics*, 57:949–961, 2025. doi: 10.1038/s41588-024-02053-6.
- Luca A. Lotta, Pawan Gulati, Felix R. Day, et al. Integrative genomic analysis implicates limited peripheral adipose storage capacity in the pathogenesis of human insulin resistance. *Nature Genetics*, 49:17–26, 2017. doi: 10.1038/ng.3714.
- Anubha Mahajan, Daniel Taliun, Matthias Thurner, et al. Fine-mapping type 2 diabetes loci to single-variant resolution using high-density imputation and islet-specific epigenome maps. *Nature Genetics*, 50:1505–1513, 2018. doi: 10.1038/s41588-018-0241-6.

- Matthew T. Maurano, Richard Humbert, Eric Rynes, Robert E. Thurman, Eric Haugen, Hao Wang, Alex P. Reynolds, Richard Sandstrom, Hongzhu Qu, Jennifer Brody, Anthony Shafer, Fidencio Neri, Kristen Lee, Tanya Kutayavin, Sandra Stehling-Sun, Audra K. Johnson, Theresa K. Canfield, Erika Giste, Morgan Diegel, Daniel Bates, R. Scott Hansen, Shane Neph, Peter J. Sabo, Shelly Heimfeld, Antony Raubitschek, Steven Ziegler, Chris Cotsapas, Nona Sotoodehnia, Ian Glass, Shamil R. Sunyaev, Rajinder Kaul, and John A. Stamatoyannopoulos. Systematic localization of common disease-associated variation in regulatory DNA. *Science*, 337(6099):1190–1195, 2012. doi: 10.1126/science.1222794.
- Lorenzo Pasquali, Kyle J. Gaulton, Santiago A. Rodríguez-Seguí, et al. Pancreatic islet enhancer clusters enriched in type 2 diabetes risk-associated variants. *Nature Genetics*, 46:136–143, 2014. doi: 10.1038/ng.2870.
- Alexander Sasse, Bernard Ng, Anna E. Spiro, Shinya Tasaki, David A. Bennett, Christopher Gaiteri, Philip L. De Jager, Maria Chikina, and Sara Mostafavi. Benchmarking of deep neural networks for predicting personal gene expression from DNA sequence highlights shortcomings. *Nature Genetics*, 55(12):2060–2064, 2023. doi: 10.1038/s41588-023-01524-6.
- Miriam S. Udler, Jaegil Kim, Marcin von Grotthuss, et al. Type 2 diabetes genetic loci informed by multi-trait associations point to disease mechanisms and subtypes: A soft clustering analysis. *PLoS Medicine*, 15(9):e1002654, 2018. doi: 10.1371/journal.pmed.1002654.
- Gao Wang, Abhishek Sarkar, Peter Carbonetto, and Matthew Stephens. A simple new approach to variable selection in regression, with application to genetic fine mapping. *Journal of the Royal Statistical Society: Series B*, 82(5):1273–1300, 2020. doi: 10.1111/rssb.12388.
- Angli Xue, Yang Wu, Zhihong Zhu, et al. Genome-wide association analyses identify 143 risk variants and putative regulatory mechanisms for type 2 diabetes. *Nature Communications*, 9: 2941, 2018. doi: 10.1038/s41467-018-04951-w.

A Trait panel

Table 3: Trait panel, data sources and pre-specified expected tissue. N is the GWAS sample size with the number of cases in parentheses for binary traits; fine-mapping regime is as reported by the Open Targets Platform (PICS around reported top hits when no summary statistics were ingested; PICS on summary statistics; SuSiE-inf, with the share of credible sets so mapped where mixed); n is the number of variants scored.

Trait	Expected	Accession	Source	N (cases)	Fine-mapping	n
Alanine aminotransferase	Liver	GCST90474304	UKB-WGS 2025	431,897	PICS (sumstats)	150
Alkaline phosphatase	Liver	GCST90018942	Sakaue 2021	463,178	PICS (sumstats)	150
Asthma	Immune	GCST010043	Han 2020	536,345 (88,486)	PICS (top hits)	122
Atopic dermatitis	Skin	GCST90503108	Oliva 2025	658,426 (56,146)	PICS (sumstats)	85
Atrial fibrillation	Heart	GCST90624411	Yuan 2025	2,584,013 (275,950)	PICS (sumstats)	150

Trait	Expected	Accession	Source	<i>N</i> (cases)	Fine-mapping	<i>n</i>
Body mass index	Brain	GCST90474606	UKB-WGS 2025	456,892	PICS (sumstats)	150
Coronary artery disease	Artery	GCST90132314	Aragam 2022	1,165,690	SuSiE-inf (98%)	150
Crohn’s disease medication (FinnGen proxy)	Immune	FINNGEN_R12_RX_ CROHN_1STLINE	FinnGen R12	500,348 (106,320)	SuSiE	50
Free cholesterol in large LDL	Liver	GCST90501150	Zoodma 2025	450,015	PICS (sumstats)	111
HDL cholesterol	Liver	GCST90501112	Zoodma 2025	450,015	PICS (sumstats)	150
Heart rate	Heart	GCST90480666	Verma 2024	609,495	PICS (sumstats)	150
Intelligence	Brain	GCST006250	Savage 2018	269,867	SuSiE-inf (95%)	150
Lung function (FEV1/FVC)	Lung	GCST90244094	Shrine 2023	588,452	PICS (top hits)	150
Platelet count	Immune	GCST90662907	Jee 2025	928,679	PICS (top hits)	150
Systolic blood pressure	Artery	GCST90000066	Surendran 2020	870,217	PICS (sumstats)	148
Triglycerides	Liver	GCST006613	Klarin 2018	297,626	PICS (top hits)	109
Type 1 diabetes	Immune	GCST90013445	Robertson 2021	59,527 (22,153)	PICS (sumstats)	66
Type 2 diabetes	Islet	GCST009379	Mahajan 2018	898,130 (74,124)	PICS (top hits)	150
Type 2 diabetes (replication)	Islet	GCST006867	Xue 2018	659,316 (62,892)	SuSiE-inf (93%)	118
Ulcerative colitis	Immune	GCST90446794	Liu 2023	375,508 (23,252)	SuSiE-inf (95%)	150

B Track census

Table 4: AlphaGenome human track census by tissue group. **Histone (5 marks)** counts only the five marks scored in this study (H3K27ac, H3K36me3, H3K4me1, H3K4me3, H3K9ac); this is the count used in the track-coverage correlations. The pancreatic islet has no ATAC tracks and a single DNase track.

Tissue	RNA-seq	ATAC	DNase	Accessibility	Histone (all)	Histone (5 marks)
Islet	6	0	1	1	5	5
Liver	16	3	4	7	27	18
Fat	12	2	2	4	13	9
Muscle	19	0	6	6	41	27
Brain	39	1	18	19	63	45
Immune	54	13	25	38	100	63
Heart	42	8	13	21	36	27
Artery	49	1	12	13	42	30
Lung	13	3	4	7	17	14
Intestine	22	5	9	14	38	30
Skin	21	1	6	7	37	26

C Superseded datasets

Table 5: Datasets of the originally assembled panel that were replaced after internal review, with their results under the same scoring. Rank, z_{exp} and P are as in Table 1. These results are superseded and are not part of any statistic in the main text.

Original dataset			Expected	n	Rank	Top	z_{exp}	P	Reason for replacement
Ulcerative colitis, PheCode 555.21 (Verma 2024, GCST90480318)			Immune	149	1	Immune	-0.01	0.99	2,312 cases; credible sets almost all single variants at marginal significance. Replaced by Liu 2023 (23,252 cases).
Type 1 diabetes with ophthalmic manifestations, PheCode 250.13 (Verma 2024, GCST90479877)			Immune	72	7	Skin	0.06	0.67	Sub-phenotype with 3,569 cases. Replaced by Robertson 2021 (22,153 cases).
Body mass index (Kichaev 2019, GCST007039)			Brain	150	3	Fat	0.10	0.80	Functionally re-weighted GWAS; credible sets built around reported top hits; no summary statistics through the GWAS Catalog, so no matched LDSC-SEG run was possible. Replaced by UK Biobank WGS 2025.
Cognitive ability / education / schizophrenia pleiotropy (Lam 2019, GCST008595)			Brain	150	2	Immune	-0.06	0.82	Pleiotropic meta-analysis retrieved under the schizophrenia term; credible sets built around reported top hits; no summary statistics through the GWAS Catalog. Replaced by Savage 2018 intelligence.